

STETTLER AGDM, AB, Canada

WMO: 712890

Lat: 52.3469N

Lon: 112.5956W

Elev: 795

StdP: 92.13

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.7	-26.0	-32.3	0.2	-28.5	-29.4	0.3	-25.9	12.2	-6.1	10.8	-6.3	3.6	190	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	29.1	17.1	27.2	16.2	25.2	15.6	18.8	25.6	17.6	24.6	16.6	23.2	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.8	22.0	15.1	11.8	20.3	14.0	11.0	19.2	56.9	25.3	52.8	24.7	49.7	23.3	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.7	8.5	DB	-32.2	32.3	2.9	2.1	-34.3	33.8	-36.1	35.0	-37.7	36.2	-39.8	37.7	(4)
(5)			WB	-32.0	21.3	3.2	1.4	-34.3	22.4	-36.2	23.2	-38.0	24.0	-40.3	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-8.8	-9.8	-3.7	3.9	10.5	14.3	17.0	15.8	11.1	4.5	-3.9	-9.7
	DBStd	11.45	7.89	7.65	7.43	5.30	4.34	3.10	2.94	3.40	4.56	5.10	6.95	7.64
	HDD10.0	3052	584	556	424	190	47	3	0	2	39	180	417	611
	HDD18.3	5447	842	789	682	433	244	124	60	91	218	428	666	869
	CDD10.0	689	0	0	1	7	62	134	218	183	74	10	1	0
	CDD18.3	43	0	0	0	0	1	5	21	14	2	0	0	0
	CDH23.3	866	0	0	1	3	54	95	344	266	100	4	0	0
	CDH26.7	214	0	0	0	0	7	14	97	69	27	0	0	0
Wind	WSAvg	4.2	4.5	4.2	4.3	4.7	4.7	4.2	3.6	3.5	3.9	4.4	4.3	4.3
Precipitation	PrecAvg	423	21	12	18	31	47	79	77	57	34	19	15	14
	PrecMax	529	63	26	38	60	85	114	166	107	77	45	56	30
	PrecMin	289	1	3	2	4	16	25	24	6	3	5	3	5
	PrecStd	71	15	7	9	16	21	26	34	24	20	9	12	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.4	14.4	22.2	27.6	28.4	31.6	31.1	29.9	23.3	13.9	7.1
		MCWB	2.6	2.6	6.8	10.3	13.4	16.3	18.3	17.2	15.5	11.9	7.0	3.3
	2%	DB	4.8	3.9	10.3	18.9	24.7	26.0	29.3	28.6	26.2	18.5	9.6	4.6
		MCWB	1.9	1.0	5.0	9.2	12.3	15.6	18.4	16.7	14.3	10.0	4.9	1.7
	5%	DB	3.3	2.2	7.2	15.7	22.3	23.8	27.2	26.3	23.2	15.3	6.4	2.5
		MCWB	0.9	0.0	3.2	7.1	11.3	14.6	17.4	16.4	13.2	8.3	2.7	0.1
	10%	DB	1.4	0.4	4.8	12.9	19.7	21.8	24.9	24.1	20.0	12.4	4.4	0.4
		MCWB	-0.6	-1.3	1.9	6.0	10.2	13.7	16.7	15.7	11.9	6.7	1.4	-1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.9	7.7	11.4	14.4	18.3	21.1	19.8	16.3	12.5	7.7	3.6
		MCDB	5.8	5.9	13.2	20.9	24.4	25.2	28.4	26.4	26.3	21.6	13.4	6.7
	2%	WB	2.1	1.2	5.1	9.6	13.1	16.7	19.4	18.2	14.9	10.5	5.1	1.9
		MCDB	4.5	3.5	9.7	17.8	22.6	23.6	26.4	25.3	24.4	17.5	8.9	4.4
	5%	WB	1.0	0.1	3.3	7.9	12.0	15.6	18.2	17.1	13.7	8.7	3.1	0.2
		MCDB	3.3	2.0	6.9	14.7	20.8	21.8	25.1	24.4	21.7	14.4	6.0	2.4
	10%	WB	-0.5	-1.2	2.0	6.4	10.9	14.4	17.2	16.1	12.5	7.2	1.7	-1.5
		MCDB	1.4	0.4	4.7	12.1	18.4	20.3	23.6	22.8	19.5	11.9	3.9	0.3
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.6	9.7	11.5	13.0	11.7	13.0	13.1	12.9	11.4	9.0	8.7
		MCDBR	9.7	10.5	12.2	17.2	17.2	15.3	17.2	17.6	18.6	15.9	11.5	9.6
		MCWBR	7.7	8.4	7.7	8.7	7.4	6.7	7.7	7.9	8.3	8.6	7.5	7.6
	5% WB	MCDBR	9.6	10.3	11.5	15.9	15.4	13.4	15.0	15.5	17.0	15.5	10.6	9.5
		MCWBR	7.7	8.4	7.7	8.6	6.8	6.7	7.7	7.7	8.0	8.5	7.4	7.7
Clear-Sky Solar Irradiance	taub	0.268	0.272	0.290	0.328	0.348	0.358	0.354	0.367	0.314	0.284	0.256	0.246	
	taud	2.284	2.374	2.428	2.366	2.360	2.366	2.407	2.349	2.469	2.531	2.464	2.322	
	Ebn at Noon	735	859	913	909	902	894	891	857	874	841	769	705	
	Edh at Noon	62	77	89	108	115	115	109	109	86	65	52	50	
All-Sky Solar Radiation	RadAvg	1.08	2.13	3.53	4.72	5.75	6.09	6.39	5.21	3.62	2.17	1.11	0.80	
	RadStd	0.10	0.18	0.27	0.44	0.36	0.39	0.33	0.36	0.39	0.25	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (86 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page